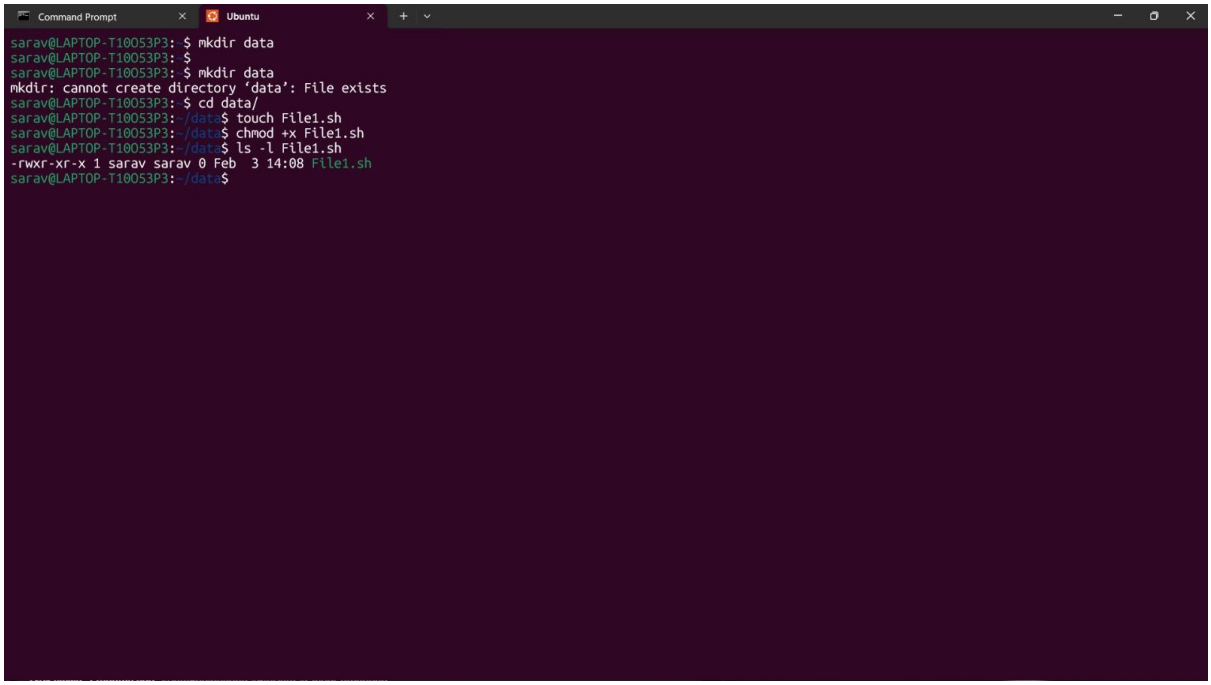
embedUR

Linux Assingment-1

Name: Saravanan D
Collage: VIT University

1) Create a file and add executable permission to all users (user, group and others)

mkdir data
cd data/
touch File1.sh
chmod +x File1.sh
ls -l File1.sh



2) Create a file and remove write permission for group user alone.

touch File2.sh
chmod g-w File2.sh
ls -l File2.sh

```
sarav@LAPTOP-T10O53P3:~/data$ touch File2.sh
sarav@LAPTOP-T10O53P3:~/data$ chmod g-w File2.sh
sarav@LAPTOP-T10O53P3:~/data$ ls -l File2.sh
-rw-r--r-- 1 sarav sarav 0 Feb  3 14:18 File2.sh
sarav@LAPTOP-T10O53P3:~/data$
```

3)Create a file and add a softlink to the file in different directory (Eg : Create a file in dir1/dir2/file and create a softlink for file inside dir1)

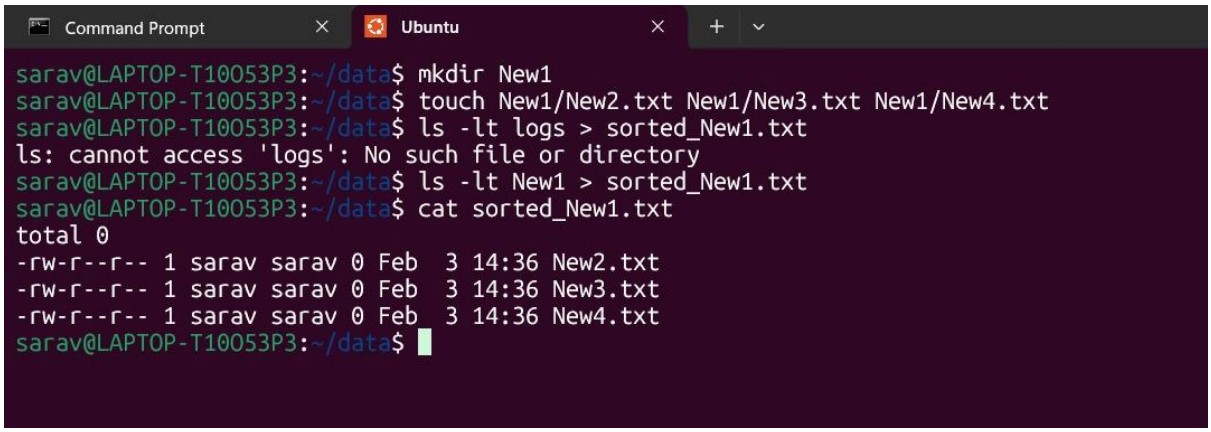
mkdir -p Dir1/Dir2
touch Dir1/Dir2/File3.sh
ln -s Dir1/Dir2/File3.sh Dir1/File3_link.sh
ln -l Dir1

```
sarav@LAPTOP-T10O53P3:~/data$ mkdir -p Dir1/Dir2
sarav@LAPTOP-T10O53P3:~/data$ touch Dir1/Dir2/File3.sh
sarav@LAPTOP-T10O53P3:~/data$ ln -s Dir1/Dir2/File3.sh Dir1/File3_link.sh
sarav@LAPTOP-T10O53P3:~/data$ ls -l Dir1
total 4
drwxr-xr-x 2 sarav sarav 4096 Feb  3 14:21 Dir2
lrwxrwxrwx 1 sarav sarav   18 Feb  3 14:23 File3_link.sh -> Dir1/Dir2/File3.sh
sarav@LAPTOP-T10O53P3:~/data$
```

4)Use ps command with options to display all active process running on the system

ps -ef

```
sarav@LAPTOP-T10O53P3:~/data$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 13:56 ?        00:00:00 /sbin/init
root           2       1  0 13:56 ?        00:00:00 /init
root           6       2  0 13:56 ?        00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          42       1  0 13:56 ?        00:00:00 /usr/lib/systemd/systemd-journald
root          96       1  0 13:56 ?        00:00:00 /usr/lib/systemd/systemd-udevd
systemd+     107       1  0 13:56 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+     108       1  0 13:56 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
root         152       1  0 13:56 ?        00:00:00 /usr/sbin/cron -f -P
message+     153       1  0 13:56 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root         160       1  0 13:56 ?        00:00:00 /usr/lib/systemd/systemd-logind
root         164       1  0 13:56 ?        00:00:00 /usr/libexec/wsl-pro-service -vv
syslog       175       1  0 13:56 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         182       1  0 13:56 hvc0     00:00:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
root         185       1  0 13:56 tty1     00:00:00 /sbin/agetty -o -p -- \u --noclear - linux
root         186       1  0 13:56 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root         259       2  0 13:56 ?        00:00:00 /init
root         261     259  0 13:56 ?        00:00:00 /init
sarav        265     261  0 13:56 pts/0    00:00:00 -bash
root         266       2  0 13:56 pts/1    00:00:00 /bin/login -f
sarav        311       1  0 13:56 ?        00:00:00 /usr/lib/systemd/systemd --user
sarav        312     311  0 13:56 ?        00:00:00 (sd-pam)
sarav        336     266  0 13:56 pts/1    00:00:00 -bash
sarav        713     265 99 14:32 pts/0    00:00:00 ps -ef
sarav@LAPTOP-T10O53P3:~/data$
```

5) Create 3 files in a dir1 and re-direct the output of list command with sorted by timestamp of the files to a file

mkdir New1
touch New1/New2.txt New1/New3.txt New1/New4.txt
ls -lt New1 > sorted_New1.txt